

Rethinking the Hidden Risk of Reranking: Achieving Risk-aware Reranking with Information Gain for RAG with LLMs

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Abstract

Retrieval-augmented generation (RAG) has become a cornerstone for enhancing large language models (LLMs) with real-time information from the Web, but its performance often heavily depends on the quality of the retrieved documents. Given that RAG systems frequently draw from vast and often noisy Web corpora, ensuring the reliability of retrieved content is paramount. While rerankers improve the factual accuracy of the RAG system by elevating the proportion of ground-truth documents (GD) in high-ranked results, the shifts of document type distributions during reranking remain unclear, hindering the understanding of the reranker's behavior. To bridge this gap, we conduct an empirical study to categorize documents and compare their distribution before and after reranking. We reveal a counterintuitive finding: though rerankers improve the proportion of GD, they also significantly increase the proportion of harmful documents (HD) in top-ranked retrieved documents. It not only narrows the potential context window for ranking the GD higher but also increases the risk of HD misleading the LLMs, potentially leading to the generation and propagation of misinformation across Web platforms. Motivated by this finding, we propose a risk-aware reranking method for RAG with LLMs, which balances the

risk and benefit during reranking. Given a query, the RAG framework first retrieves relevant documents. Then, our approach quantifies the potential beneficial and harmful impacts of various documents on the LLMs' generation. To estimate the impacts, we conduct a dual-aspect document impact assessment via information gain, which employs a risk clipping to avoid the numerical fluctuations in the estimation. Finally, we conduct the reranking according to the potential impact of each document, enabling the reranker to significantly reduce the HD proportion. Experiments and analysis across multiple models and datasets, including Wikipedia, web news, and research papers, show the effectiveness of our method. Our code is available at https://github.com/lzz335/hidden_risk_of_reranking.

CCS Concepts

• Information systems → Retrieval models and ranking.

Keywords

Large Language Models, Retrieval-augmented Generation, Reranker

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1 Introduction

Retrieval-augmented Generation (RAG) is a critical technique for modern Web search and question-answering systems, enhancing

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the robustness and factual accuracy of large language models (LLMs) by incorporating up-to-date knowledge retrieved from the Web. Critically, the performance of RAG systems is highly dependent on the quality of the retrieval process [7, 8, 13, 36, 71]. Specifically, the factual mistakes or misleading information within retrieved documents may undermine the factual accuracy of the LLM generation [2, 17, 35, 53, 56, 70], ultimately degrading the quality and trustworthiness of information delivered to Web users. To address these limitations, some researchers optimize the retrieval process to enhance the reliability and relevance of retrieved documents [42, 55].

Current research primarily focuses on two key lines to improve the quality of retrieved documents. (1) Some researchers improve the quality of the retrieval process. They precompute document semantic vectors and use MIPS or cosine similarity to measure the similarity between the query and documents [40, 45]. (2) Further, some researchers propose to append a fine-grained reranker after the retrieval stage to rerank the retrieved documents [1, 28, 66]. Unlike pre-computed embeddings in retrieval, these methods typically generate adaptive document representations considering each specific query¹ [1, 66]. This produces query-aware document representations that enhance semantic interactions between documents and queries, consequently improving their relevance [7, 25, 71].

To enhance the reranker, researchers attempt to use the following three categories of approaches. (1) *listwise reranking* [28, 38] considers the whole candidate list and directly outputs a ranking list for all documents. As the candidate set tends to contain a great number of long documents, its computational overhead is quite high. For example, LLM-based listwise ranking [7, 71] requires feeding all candidate documents into its limited context window. (2) *pairwise reranking* [29, 59] compares every two candidate documents and then performs a bubble sort to obtain the final list. The computational complexity also reaches $O(|D|^2)$ for a candidate set with $|D|$ documents. (3) *pointwise reranking* [1, 66] is an efficient and widely-used strategy that assigns a relevance score for each document-query pair and then ranks all pairs according to their scores. Current rerankers primarily focus on semantic similarity during reranking; however, recent studies found that semantically related yet unsupportive documents distract LLMs from correct answers [4, 54]. Although rerankers enhance the overall performance of RAG by improving the proportion of ground-truth documents in high-ranked results, it remains unexplored how the distribution of distracting documents shifts during reranking. To bridge this gap, we conduct an empirical study to investigate the distribution shifts of different categories of documents during the reranking process. Following previous work [4, 54], we first categorize the retrieved documents into three types (see examples in Fig. 1, App. C.2.1): (1) ground-truth document (GD), which means the document leads the LLM to generate the ground-truth answer; (2) mediocre document (MD), which refers to completely irrelevant document that does not interfere with LLMs' generation; (3) harmful document (HD), which refers to document with fake information or topic-similar interference, which might introduce misinformation and raise the risk of incorrect LLM output. Our study finds that **(1) while rerankers improve the proportion of GD to enhance the factuality of**

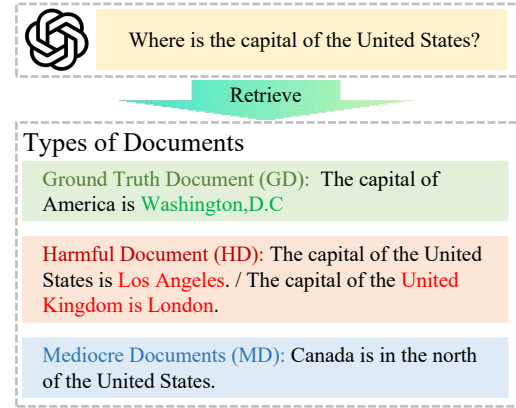


Figure 1: Examples of GD, MD, and HD. GD is the ground-truth document, but HD might interfere with the LLM generation process.

LLM generation, they also increase the probability of HD interfering, potentially constraining the RAG system from achieving better performance; (2) **this challenge with HDs is distinct from a fact-checking task, because the majority of these documents are factually correct. Rather, they are misleading due to their superficial topical similarity, which can mislead the model's reasoning even without containing explicit falsehoods.** Based on our findings, we suggest that maintaining the recall rate of GD while reducing the incidence of HD may further improve the overall performance of RAG systems.

In this paper, to prevent an undesirable increase in the rate of HD after reranking, we propose to redesign the objective of the reranker with a utility function that balances the risks and benefits of reranking, considering information gain. Specifically, given a query, the RAG pipeline first retrieves documents from the database. Then, we design a dual-aspect document impact assessment module that measures the beneficial (or harmful) impact of the retrieved document to quantify its benefits (or risks) with information gain. To anticipate the impact of reranking, we employ a utility estimation module that (1) estimates the potential impact of the document on LLMs' generation, and (2) balances the beneficial and harmful impact to obtain the utility score of the document. Additionally, we propose a risk clipping strategy to avoid numerical fluctuations in the utility estimation. Finally, we rerank documents according to the values of the utility score and prompt the LLM to generate a response considering the query and the reranked documents.

Experiments demonstrate our reranker substantially outperforms existing methods across two open-source and one proprietary LLMs on three cross-domain datasets, achieving state-of-the-art performance. Our contributions are as follows: (1) Following the previous works about the definition of the harmful document in RAG [4, 54], we are the first to discover the counterintuitive finding that reranking in RAG also increases the proportion of harmful documents presented to LLMs, rather than solely increasing the proportion of ground-truth documents. (We also provide a potential explanation for our finding in Sec 3.2). (2) We redesign the optimization objective of the reranker in RAG by introducing a novel

¹The most common approach is to feed each query and document pair into an encoder for representations. (i.e. Bert [5]).

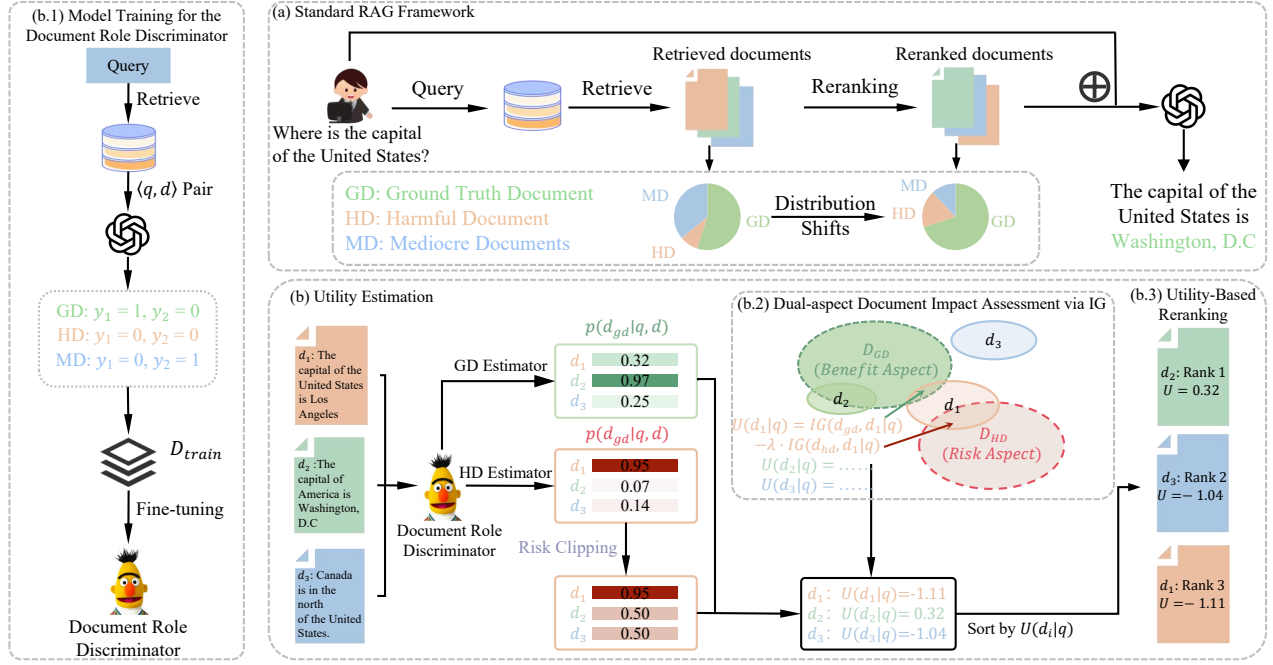


Figure 2: The overview of our work. We first conduct an empirical study and find that the reranker improves the proportion of both GD and HD, simultaneously (the pie charts). Our work focuses on the reranker in the RAG framework (a). We first construct a dataset D_{train} and train our Document Role Discriminator (b.1). For the query “Where is the capital of the United States?”, we assess the retrieved documents’ dual-aspect impact via information gain (b.2). Then we use the Discriminator to estimate the information gains, and calculate the utility $U(d|q)$ of each document (b). Utility-Based Reranking reranks the documents according to the value of utility (b.3). Finally, the RAG system generates the answer based on the query and the reranked documents.

utility function that balances the beneficial and harmful impact of retrieved documents using dual-aspect information gain. (3) Our approach is plug-and-play with mainstream RAG frameworks and achieves better performance across three baseline models and four datasets.

2 Related Work

2.1 Retrieval-augmented Generation

RAG enhances LLMs by retrieving relevant external text. Currently, researchers primarily focus on two main lines to improve retrieved passage quality:

2.1.1 Retriever. Retriever rapidly extracts highly relevant documents from massive databases [7, 71]. Robertson et al. [41] introduced simple lexical matching methods to retrieve the most relevant documents. Reimers and Gurevych [40] propose using cosine similarity between sentence embeddings to better evaluate semantic similarities. Wang Yuxin [52] uses large-scale hybrid embedding techniques to generate more comprehensive and precise word vector representations. Xiao et al. [55] leverages the RetroMAE pre-training algorithm to improve the model’s semantic representation. Chen et al. [1] combines previous works and expands the corpus to support multilingual scenarios. Meanwhile, some researchers explore hybrid approaches that combine existing vector methods

with traditional lexical matching to improve retriever effectiveness [42, 60].

2.1.2 Reranker. Reranker employs supervised training on annotated datasets, such as MTEB [32], to increase the proportion of ground-truth documents after reranking [7, 8, 71]. Nogueira and Cho [33] introduces fine-tuned BERT models to capture relevance between sentence pairs. Research proposes contrastive learning to enhance performance [1, 66] while substituting pairwise loss for cross-entropy loss [56]. Chen et al. [2] further optimizes inference efficiency through hierarchical self-distillation strategies. Liu et al. [28] processes all candidate documents simultaneously to acquire global awareness. Meanwhile, several methods enable large language models to directly output ranked lists or relevance scores [30, 37, 62].

2.2 LLM Hallucination in RAG

Hallucination refers to the generation of content by LLMs that contradicts verifiable real-world facts or the user input context [15, 68]. In RAG systems, knowledge conflicts constitute a primary cause of hallucinations [58]. Some researchers aim to enhance the accuracy of LLMs by increasing their dependency on external knowledge [16, 63], such as through fine-tuning [26] or decoder modification [44, 50]. However, external documents often contain substantial noise or deliberately fabricated misinformation, which complicates

precise processing and response [8, 58]. Hsu et al. [14] develops a conflict resolution method that analyzes the probability of contradiction across documents. Some researchers propose an integrated fact-checking framework combining model outputs with external tools to detect factual errors [3] or verify the factivity of documents using LLMs [51]. Recent studies have further investigated the impact of Inter-Context conflicts on RAG performance [31, 58]. Cucanasu et al. [4] suggests irrelevant knowledge sometimes improves system outcomes. Wu et al. [54] supposes that seemingly relevant content often impairs the quality of generation, and LeVine and Varjavand [23] observes that recall-oriented approaches create information bottlenecks.

Our method enhances document quality at the reranking stage by lowering the HD to resolve external knowledge conflicts, thus improving RAG system performance.

3 Empirical Study

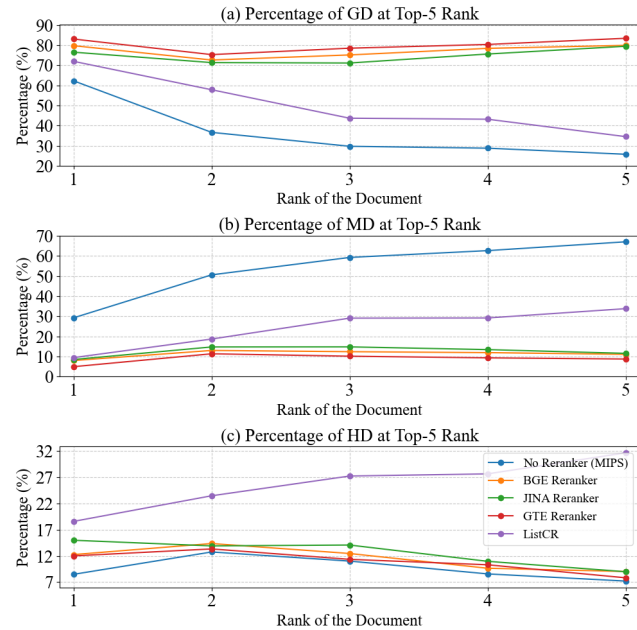


Figure 3: The percentages of HD, MD, and GD among top-5 results. The blue line indicates the results without reranking. The reranker (BGE, JINA, GTE and ListCR) significantly increases the percentage of GD (a) while decreasing that of MD (b). Note that the percentage of HD exhibits an increase (c).

We investigate how existing rerankers affect the distribution of top-ranked documents. Our analysis shows that while reranking improves the proportion of GD, it also increases the proportion of HD.

3.1 Experimental Settings

We evaluate four widely adopted reranker models: bge-reranker-v2-m3 [1], jina-reranker-v2-base-multilingual [9], gte-multilingual-reranker-base [66], and ListConRanker [28] on the MuSiQue [49]

dataset. We use bge-large-en as the retriever [57, 64, 65]. Following the typical RAG workflow, given a query, we first calculate its embedding with the retriever and then select the top 50 candidate documents using MIPS (Maximum InnerProduct Search). Subsequently, each reranker model computes relevance scores for these retrieved passages, and we rank the candidate passages accordingly based on the scores.

3.2 RQ: How Does Reranking Affect the Distribution of High-ranked Retrieved Documents?

We investigate how the reranking affects the distribution of the top-ranked retrieved documents. First, we classify the documents in the dataset into positive and negative examples based on whether the document leads the LLM to generate the ground-truth answer. Subsequently, following existing work [4, 54], due to misclassification by the MIPS or reranker, we further divide the high-ranked negative examples (False Positive, FP) into (1) Mediocre Documents (MD): completely irrelevant documents, which do not interfere with LLMs’ generation; (2) Harmful Documents (HD): documents containing false information or interference similar to topics, can introduce misinformation and increase the risk of incorrect LLM output. Finally, we use GPT-4o [34] to identify the types of FP. We compare the distribution of false positives within the top five retrieved documents between before reranking and after reranking via MIPS in Fig. 3.

Existing studies [4, 54] find that: (1) Increasing the proportion of GD enhances RAG system performance; (2) Increasing the proportion of HD diminishes RAG system performance. Given the empirical results, we find that **though the reranker can improve the proportion of ground-truth documents, it also significantly improves the proportion of harmful documents** (significant under the McNemar’s Test with $p < 0.05$, see details in App. D). To ensure this finding is not an artifact of a specific retrieval model, we conduct further experiments with other SOTA retrievers and find the same trend (see details in App. E). Specifically, we further analyze the distribution of HDs. We find that documents exhibiting only lexical or syntactic similarity constitute 88.7% of these high-ranked HDs, while those containing factual errors account for **only 11%**. This implies that mitigating high-ranked HDs cannot be solved merely by adding a fact-checking module after reranking. Instead, we frame this as an optimization problem within the reranking process itself.

Furthermore, this observation leads to a crucial question: **why does the proportion of HD increase in such a counterintuitive manner?** We argue that the limited capacity of rerankers forces them to adopt “shortcuts” [5, 18, 43]—relying on locally correlated, superficial features instead of complex global semantic reasoning [10, 12, 18, 20, 27, 47]. To provide direct evidence for this, our analysis of the reranker’s attention mechanism in Fig. 4 confirms this behavior. The heatmaps show that even in deep layers, the model’s focus remains on simple lexical matching of identical or adjacent tokens. This over-reliance on superficial cues like shared entities and keyword co-occurrence is the exact vulnerability that many harmful documents exploit. By being topically relevant and sharing keywords with the query, they “trick” this shortcut-based

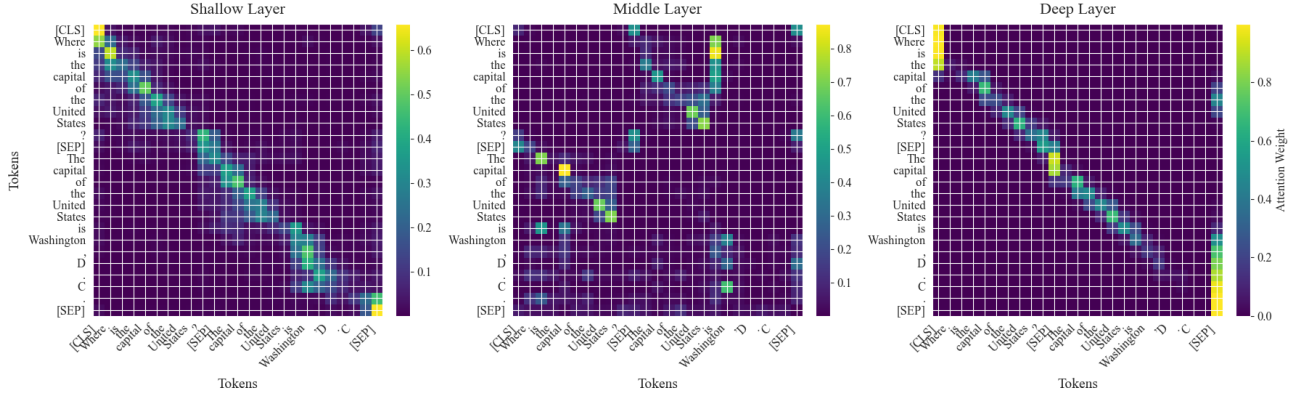


Figure 4: Heatmaps of the GTE reranker’s attention matrices, progressing from shallower (left) to deeper (right) layers. Although GTE is a cross-encoder model, its attention primarily focuses on adjacent and identical tokens.

mechanism. Therefore, we argue that the model’s limited capacity forces a reliance on superficial features, introducing a critical vulnerability. Harmful documents that mimic these features can exploit this vulnerability, leading to their promotion and a subsequent increase in the HD proportion after reranking.

According to Cuconasu et al. [4] and Wu et al. [54], HD can distract LLMs or introduce harmful perturbations, leading to sub-optimal performance. However, few researchers focus on **the increase of HD after reranking**. This might lead to unsatisfactory performance of RAG, and solving this problem may provide an opportunity to further enhance the overall performance of RAG.

4 Method

4.1 Overview

We first construct a standard RAG framework for LLMs (Sec. 4.2). Then, we construct a BERT-based reranker model to balance risk and benefit from retrieved documents, where the reranker is integrated into the RAG. Our work focuses on refining the reranker within the RAG framework and consists of three modules: (1) **Dual-aspect Document Impact Assessment** (Sec. 4.3) quantifies document risks (or benefits) by assessing their beneficial (or harmful) contributions to LLM outputs using information gain. (2) **Utility Estimation** (Sec. 4.4) learns to estimate the potential utility of the document on LLMs’ outputs, which balances the risks and benefits of the document obtained in Sec. 4.3. It also employs a risk clipping to avoid the numerical fluctuations in the estimation of the utility function. (3) **Utility-Based Reranking** (Sec. 4.5) sorts documents in descending order according to the utility function values.

4.2 Standard RAG Framework

We formalize the standard RAG framework. Given a query q , the RAG system first retrieves m candidate documents $\mathcal{D} = \{d_1, d_2, \dots, d_m\}$ from a database using a retriever. For each retrieved document $d_i \in \mathcal{D}$, a reranker employs a scoring function $g(q, d_i) : q \times d_i \rightarrow S_i$ to evaluate its relevance to q . The RAG framework follows $\hat{\mathcal{D}} =$

$\operatorname{argmax}_{|\mathcal{J}|=k} \sum_{j \in \mathcal{J}} S_j$ to construct a set of the top- k most relevant documents $\hat{\mathcal{D}}$. Finally, the RAG framework employs an LLM to generate an answer based on the q and $\hat{\mathcal{D}}$.

4.3 Dual-aspect Document Impact Assessment via Information Gain

Based on information gain, we assess the risk and benefit of the retrieved documents by estimating the beneficial and harmful contributions of the documents to LLM outputs. Given a query q and retrieved document d , following Federici et al. [6] and Zhu et al. [69], the information gain $IG(d_{gd}, d|q)$ can quantify d ’s contribution to the LLM generating ground-truth outputs, where the d_{gd} is the distribution of ground-truth documents. We use d_{gd} to quantify the benefit of the document d . Similarly, since d_{hd} represents a distribution of harmful documents that impairs LLM generation, information gain $IG(d_{hd}, d|q)$ quantifies document d ’s impact in producing incorrect answers, which can quantify the risk of the document d .

Following Quinlan [39], we use mutual information to quantify information gain. We first express the $IG(d_{gd}, d_j|q)$ as an integral in terms of mutual information. We then apply the chain rule of probability to decompose $p(d_j, d_{gd} | q)$, leading to the expectation-based formulation of information gain so that we can train a model to estimate it:

$$IG(d_{gd}, d_j|q) = \mathbb{E}_{p(d_j, d_{gd}|q)} [\log p(d_{gd} | q, d_j) - \log p(d_{gd} | q)] \quad (1)$$

Similarly, we have $IG(d_{hd}, d_j|q) = \mathbb{E}_{p(d_j, d_{hd}|q)} [\log p(d_{hd} | q, d_j) - \log p(d_{hd} | q)]$.

4.4 Utility Estimation

To coordinate the benefits and risks of retrieved documents, we propose a utility function grounded in dual-aspect information gain.

4.4.1 Step 1. Formalization of the Utility Function. Empirical analysis reveals that while reranking improves RAG performance by elevating the rankings of good documents (GD), it simultaneously increases the prevalence of harmful documents (HD). Current rerankers prioritize positive GD contributions while overlooking HD’s detrimental effects, resulting in suboptimal outcomes. Consequently, we redesign the reranking objectives to simultaneously evaluate the benefits of GDs and risks of HDs, ensuring higher GD proportions while suppressing HD. We employ the Markowitz model [24] to coordinate the benefits and risks of retrieved documents during reranking. The Markowitz model maximizes expected rewards for a given level of risk. It provides a practical approach to reduce overall risk while enhancing the potential reward. This framework quantifies the utility of a document through the function $U(d) = I(d) - \lambda \cdot R(d)$. $I(d)$ represents beneficial contributions of d to LLM outputs while $R(d)$ denotes harmful contributions. The coefficient λ controls risk aversion intensity (see discussion of λ in App. F). Thus, building on the work in Sec. 4.2, we find that $IG(d_{gd}, d_j|q)$ effectively quantifies document d ’s contribution to generating correct answers. Therefore, we employ $IG(d_{gd}, d_j|q)$ to represent the reward $I(d)$. Similarly, we utilize $IG(d_{hd}, d_j|q)$ as the risk measure $R(d)$, then we have:

$$U(d_j|q) = \underbrace{IG(d_{gd}, d_j|q)}_{\text{reward}} - \lambda \cdot \underbrace{IG(d_{hd}, d_j|q)}_{\text{risk}} \quad (2)$$

We substitute Eq. 1 into the $U(d_j|q)$. Since $\log p(d_{gd} | q)$ is constant for a fixed query q , we omit this term during reranking: $IG(d_{gd}, d_j|q) \approx \mathbb{E}_{p(d_j, d_{gd}|q)} \log p(d_{gd} | q, d_j)$. Similarly, for $IG(d_{hd}, d_j|q)$, $IG(d_{hd}, d_j|q) \approx \mathbb{E}_{p(d_j, d_{hd}|q)} (\log p(d_{hd} | q, d_j))$. After omitting constant terms independent of d_j , we estimate document d_j ’s utility under query q using Eq. 3:

$$U'(d_j | q) = \mathbb{E}_{p(d_j, d_{gd}|q)} (\log p(d_{gd} | q, d_j)) - \lambda \cdot \mathbb{E}_{p(d_j, d_{hd}|q)} (\log p(d_{hd} | q, d_j)) \quad (3)$$

4.4.1.1 Risk Clipping. To ensure the numerical stability of the utility function, we truncate $\log p(d_{hd} | q, d_j)$. Since the derivative of $\log x$ approaches infinity when x approaches 0, minor fluctuations in $p(d_{hd} | q, d_j)$ cause substantial changes in its logarithmic value when $p(d_{hd} | q, d_j) \rightarrow 0$. During reranking, an irrelevant document ideally receives a lower reranking score than that of GD. However, the logarithmic property introduces numerical instability for harmless documents since $p(d_{hd} | q, d_j) \rightarrow 0$, whereby slight variations in this probability may induce significant fluctuations in $U'(d_j|q)$. Therefore, we truncate $p(d_{hd} | q, d_j)$ and substitute it with $p'(d_{hd} | q, d_j)$ (see the discussion of clipping in App. G):

$$p'(d_{hd} | q, d_j) = \max\{p(d_{hd} | q, d_j), 0.5\}$$

4.4.2 Step 2. Utility Estimation via Document Role Discriminator (DRD). We propose to estimate the utility function U' by training a probability estimator based on fine-tuning a pre-trained model.

4.4.2.1 Dataset Construction for Training a DRD. For each collected query-document pair $\langle q, d \rangle$, we annotate two labels: (1) y_1

indicating whether document d is a GD for query q , and (2) y_2 denoting the error type of d (with $y_2 = 1$ indicating HD, and $y_2 = 0$ indicating MD) when d does not conclude the ground-truth answer. We annotate the labels using GPT-4o [34] with few-shot examples. To ensure the reliability of our annotation process, we measure the agreement between GPT-4o and a human expert using Cohen’s Kappa. The resulting coefficient of 0.814 indicates an “Almost Perfect” level of agreement, confirming the high quality of our LLM-based annotations (see Sec. 5.4 for details). We define our dataset as: $D_{\text{train}} = \{(\langle q, d \rangle, y_1, y_2) \mid q \in \mathcal{Q}, d \in \mathcal{D}, y_1 \in \{0, 1\}, y_2 \in C_{\text{error}}\}$, where $\langle q, d \rangle$ represents a query-document pair (details in App. C, examples for dataset can be found in our anonymous repository).

4.4.2.2 Model Training for the DRD. We fine-tune the DeBERTa-v3-base² [11] model for probability estimation. First, we concatenate query q and document d into the input sequence $Seq(q, d) = [q; [\text{SEP}]; d]$ as model input. The pre-trained model $f_\theta(\cdot)$ outputs its final hidden state at [CLS] position as the joint embedding vector for $Seq(q, d)$: $f_\theta(Seq(q, d)) \in \mathbb{R}^h$, where h denotes the hidden layer dimension. We then introduce two classification heads FFN_{gd} and FFN_{hd} . Head FFN_{gd} estimates $p(d_{gd} | q, d_j)$ using y_1 . It determines whether document d is GD for query q : $\hat{p}(d_{gd} | q, d) = \text{softmax}(FFN_{gd}(f_\theta(Seq(q, d))))_1$. Similarly, head FFN_{hd} estimates $p(d_{hd} | q, d_j)$ using labels y_2 . It identifies whether d is HD for q : $\hat{p}(d_{hd} | q, d) = \text{softmax}(FFN_{hd}(f_\theta(Seq(q, d))))_1$. The training objective is to minimize the cross-entropy loss $\mathcal{L}$.

$$\mathcal{L} = \mathbb{E}_{(q, d, y_1, y_2) \sim \mathcal{D}} [-\log \hat{p}(d_{gd} | q, d)^{y_1} (1 - \hat{p}(d_{gd} | q, d))^{1-y_1} - \log \hat{p}(d_{hd} | q, d)^{y_2} (1 - \hat{p}(d_{hd} | q, d))^{1-y_2}] \quad (4)$$

In summary, we first formalize the utility function in Step 1 and subsequently construct a document role discriminator in Step 2 to estimate the utility.

4.5 Utility-Based Reranking

Finally, we adopt the utility function value as the document’s final score according to Eq. 3. According to Eq. 5, we then select the top- k documents with the highest utility values to obtain reference documents; and then, those documents act as the retrieved results to augment LLMs’ text generation.

$$\hat{\mathcal{D}} = \underset{\substack{\mathcal{J} \subseteq \{1, \dots, m\} \\ |\mathcal{J}|=k}}{\text{argmax}} \sum_{j \in \mathcal{J}} U'(d_j|q) \quad (5)$$

Generally, given a query q , the RAG framework (Sec. 4.2) first retrieves relevant documents. Then, using the dual-aspect information gain principle (Sec. 4.3), we estimate the utility function of each document (Sec. 4.4). We select the top- k documents $\hat{\mathcal{D}}$ with the maximal utility values (Sec. 4.4). Finally, the LLM generates an answer based on q and $\hat{\mathcal{D}}$.

5 Experiment

5.1 Implementation Details

5.1.1 Setting. We perform model training and inference on a single NVIDIA RTX 4090 GPU using PyTorch 2.7.0 and CUDA 12.8. For fine-tuning, we use a learning rate of 1×10^{-5} and train for eight epochs, with a batch size of 8, based on the TriviaQA and

²<https://huggingface.co/microsoft/deberta-v3-base>

Table 1: The performance (Acc, EM, F1) of different methods in three models. The second column shows the baselines we compare with Ours, and the third column shows the parameters of different baselines. The bolded results indicate the best-performing method. Our improvements are significant under the McNemar’s Test with $p < 0.05$ (see details in App. D).

Method	Parameters	MuSiQue			NQ			MultiHopQA			PubMedQ-L			
		Acc	EM	F1	Acc	EM	F1	Acc	EM	F1	Acc	EM	Micro-F1	
GPT4o-mini	Vanilla	0	6.47	5.44	0.144	27.6	24.9	0.372	61.5	60.8	0.612	61.5	-	0.282
	MIPS	335M	15.4	14.7	0.235	37.7	30.2	0.447	66.8	64.7	0.659	64.2	-	0.312
	BGE	335M+560M	18.7	18.5	0.280	40.0	31.9	0.468	69.6	67.8	0.688	65.7	-	0.318
	JINA	335M+278M	18.4	16.4	0.253	38.8	31.0	0.454	68.0	65.5	0.669	66.0	-	0.327
	GTE	335M+304M	18.8	18.3	0.274	39.8	31.6	0.466	68.6	67.2	0.679	65.5	-	0.333
	ListCR	335M+326M	17.2	15.4	0.268	33.8	27.4	0.408	67.3	65.5	0.658	64.5	-	0.236
	Ours	335M+139M	23.6	23.0	0.338	41.6	32.8	0.524	70.3	68.4	0.695	66.0	-	0.350
gemma (4B)	Vanilla	0	1.07	1.02	0.038	9.73	8.94	0.145	35.7	35.5	0.359	59.6	-	0.302
	MIPS	335M	6.99	6.62	0.123	32.0	26.3	0.406	52.8	52.5	0.553	54.1	-	0.410
	BGE	335M+560M	11.2	10.5	0.171	34.4	26.3	0.410	57.5	57.4	0.598	58.6	-	0.434
	JINA	335M+278M	9.34	8.82	0.147	33.6	26.3	0.407	54.9	54.6	0.578	58.6	-	0.426
	GTE	335M+304M	10.7	10.1	0.166	33.1	26.7	0.410	55.4	54.8	0.559	59.0	-	0.430
	ListCR	335M+326M	11.3	10.5	0.188	33.8	26.4	0.403	57.8	57.8	0.598	58.6	-	0.379
	Ours	335M+139M	13.9	12.9	0.215	34.9	27.2	0.415	60.8	59.7	0.614	58.9	-	0.457
Qwen3 (8B)	Vanilla	0	6.83	5.26	0.123	32.3	14.5	0.254	54.8	49.6	0.515	61.0	-	0.267
	MIPS	335M	18.5	9.97	0.146	51.1	27.8	0.424	78.1	73.7	0.751	63.3	-	0.308
	BGE	335M+560M	21.8	12.1	0.174	53.9	29.3	0.449	81.7	78.8	0.801	66.5	-	0.412
	JINA	335M+278M	20.1	11.1	0.159	54.1	29.3	0.439	78.9	75.2	0.767	65.1	-	0.404
	GTE	335M+304M	21.8	12.3	0.176	52.4	29.1	0.437	81.5	77.6	0.789	66.0	-	0.378
	ListCR	335M+326M	19.4	10.7	0.166	53.4	28.7	0.428	80.9	76.9	0.774	64.3	-	0.350
	Ours	335M+139M	29.5	17.5	0.240	54.7	29.8	0.452	82.9	79.6	0.814	66.5	-	0.429

MuSiQue datasets (see App. C.2 for details on how we construct the dataset). We report the DRD’s performance from an IR perspective in App A. The RAG pipeline comprises retrieval, reranking, and inference stages [8]. For retrieval, we use the ChromaDB framework with the bge-large-en model to embed documents and queries [1]. Our embedding process follows the default prompt in the BAAI HuggingFace repository. All baselines for reranking use standard configurations. We set the hyperparameter $\lambda = 0.5$ in our approach (see discussion in App. F). We use a zero-shot RAG template with the [Context] + [Question] as a prompt structure (see details in our github repository). Documents are arranged by the ascending relevance scores of the top five reranked documents to reinforce the effectiveness of reranking [4, 61]. All LLMs inferences use default parameters.

5.1.2 Dataset. We conduct our experiments on the following four datasets, covering both single-hop and multi-hop problems, as well as diverse document types including web information, Wikipedia, and academic papers: MuSiQue [49], Natural Questions (NQ) [21], MultiHopQA [48], and PQA [19]. We evaluate our method on the entire MultiHopQA dataset and PQA-L (labeled), while the other datasets are test sets. Details can be found in the App. C.1.

5.1.3 Baseline. We benchmark our approach against leading SOTA open source models: BGE [1], JINA [9], GTE [66], and a listwise SOTA baseline ListCR [28]. We also evaluate two baselines: MIPS-based retrieval without rerankers (MIPS), and direct prompt-based responses from LLM without external knowledge (Vanilla).

5.1.4 Metric. Following LeVine and Varjavand [23] and our empirical study, solely evaluating reranker performance may not adequately capture its impact on the overall RAG system. Thus, consistent with Wang et al. [51], Yue et al. [61], we evaluate the complete RAG system performance. We compute accuracy (Acc) by evaluating the matching of output and gold answer using GPT-4o-mini [34]. To align with some researchers’ evaluation [22, 51, 61], we report both Extract Match (EM) and F1 scores. EM determines whether the system output completely contains the gold answer. We compute the F1 score using ROUGE-L, which measures accuracy and recall between the RAG output and the gold answer. Notably, since the PQA-L dataset consists of a three-way classification task, we use Micro-F1 instead of the standard F1 score.

5.2 Overall Performance

Following the RAG pipeline in Sec. 5.1, we compare our method with existing baselines across three datasets. We examine three benchmark LLMs representing mainstream black-box, open-source white-box, and reasoning LLMs. Tb. 1 shows that our method achieves the best performance across all evaluations.

The Vanilla method shows the worst performance, confirming that retrieval can enhance model outcomes. Similarly, MIPS yields the second-worst results, indicating that rerankers enhance top-document quality and overall performance. Unlike other rerankers that focus solely on document relevance, our reranker addresses document risk and benefits from the utility function, reranking the document accordingly. Compared to the above reranker baselines, our method achieves average improvements of 5.9 ± 2.1 , 2.0 ± 1.1 ,

Table 2: Ablation study on the key components in our method. The table values show the reduction of Acc after ablation (compared to our methods). $-I(d)$ ignores $p(d_{gd} | q, d)$ in U' , and $-R(d)$ ignores $p'(d_{hd} | q, d)$. $-Clip$ means using $p(d_{hd} | q, d)$ instad of $p'(d_{hd} | q, d)$.

Model	Components	MuSiQue	NQ	MultiHopQA	PQA-L
GPT4o-mini	$-I(d)$	-7.30	-8.08	-7.58	-2.4
	$-R(d)$	-0.98	-1.28	-2.08	-0.2
	$-Clip$	-0.25	-1.21	-1.31	-1.0
gemma (4B)	$-I(d)$	-5.67	-10.6	-7.72	+2.7
	$-R(d)$	-0.86	-1.69	-3.08	-1.1
	$-Clip$	-0.16	-0.75	-2.30	-0.7
Qwen3 (8B)	$-I(d)$	-6.24	-9.31	-9.58	-3.3
	$-R(d)$	-0.83	-6.22	-6.13	-1.3
	$-Clip$	-0.36	-4.17	-4.01	-5.2

2.8 ± 1.1 , and 0.36 ± 0.55 points on the Musique, NQ, MultiHopQA, and PQA-L datasets, respectively, and achieves better performance across almost all LLMs and datasets except the PQA-L on gemma 4B ($p_{\max} \approx 0.0248$, see App. D). While our method’s improvement in ‘Acc’ on PQA-L appears marginal in some cases, this observation can be explained by the dataset’s severe class imbalance. Specifically, the ‘yes’ label constitutes the overwhelming majority, with the distribution of the three classes being approximately 5:3:1. This makes ‘Acc’ a less informative metric, as always predicting the majority class can spuriously achieve a high score. Therefore, Micro-F1 offers a more robust and meaningful evaluation of a model’s true performance, and the significant improvement of our method in Micro-F1 demonstrates the effectiveness of our method. Additionally, our reranker has only 139M parameters, significantly smaller than those of the baseline models (278M to 560M), see the discussion of computational cost in App. B.

5.3 Ablation Study

We conduct an ablation study on our proposed method to verify the importance of each component in improving RAG’s performance. As shown in Tb. 2, $-I(d)$ indicates the omission of $p(d_{gd} | q, d)$ in the reranking utility computation. The $-I(d)$ strategy achieves the worst performance. $-R(d)$ indicates omitting $p'(d_{hd} | q, d)$ in the utility computation. The $-R(d)$ strategy also demonstrates performance degradation. We will further analyze the reasons for performance degradation after the ablation of $I(d)$ and $R(d)$ in App. A. The $-Clip$ disables clipping values of $p(d_{hd} | q, d)$. The performance degradation suggests that risk clipping maintains numerical stability in the utility function when handling harmless documents, thereby improving the accuracy of the LLM.

5.4 The Kappa Coefficient with Human Evaluators

To validate the reliability of using LLMs for annotation, we employ Cohen’s Kappa, a statistical measure that quantifies the agreement between two raters (e.g., an LLM and a human). Unlike simple accuracy, Kappa accounts for agreement that occurs by chance, providing a more robust assessment. A Kappa value exceeding

0.8 is conventionally interpreted as “almost perfect agreement,” indicating high reliability.

(a) Agreement between human and GPT-4o annotations (N=1000). Cohen’s Kappa = 0.814.

	GPT-4o (T)	GPT-4o (F)	Sum
Human Evaluator (T)	403	54	457
Human Evaluator (F)	38	505	543
Sum	441	559	1000

(b) Inter-model agreement between GPT-4o and Gemini 2.5 Flash. Cohen’s Kappa = 0.821.

	GPT-4o (T)	GPT-4o (F)	Sum
Gemini 2.5 Flash (T)	44680	5093	49773
Gemini 2.5 Flash (F)	4797	57181	61978
Sum	49477	62274	111751

For our primary evaluation, we randomly sample 1,000 instances from the dataset and compare the annotations generated by GPT-4o against those from a human expert. As shown in the agreement matrix (Tb. 3a), this comparison yields a Kappa coefficient of 0.814, confirming a strong concordance between the LLM and human annotators. To further ensure the stability of our annotation process, we conduct a secondary analysis on inter-model consistency. We compare the annotations produced by GPT-4o and Gemini 2.5 Flash across the entire dataset. The inter-model agreement matrix (Tb. 3b) shows a Kappa coefficient of 0.821. This high value suggests that, despite minor variations, the outputs from different state-of-the-art LLMs or humans are highly consistent. Taken together, these results demonstrate that our LLM-based annotations are both statistically robust and acceptable.

6 Conclusion

In this paper, we investigate whether the reranker in RAG has adverse effects on LLMs. Our empirical study reveals that conventional rerankers inadvertently increase the proportion of harmful documents (HD) in top-ranked documents while improving ground-truth documents (GD), potentially constraining the upper performance limit of the RAG-based LLMs. To address this, we propose a risk-aware utility function that quantifies both beneficial and harmful impacts of documents via dual-aspect information gain, and our model reranks the documents based on the value of the utility function. It reduces the percentage of HD while maintaining the GD percentage, thereby offering a novel perspective for reranking objectives, which enhances the performance of RAG in LLMs. Experiments across three LLMs and four datasets demonstrate the method’s effectiveness.

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A Analysis of $I(d)$ and $R(d)$

Table 4: Ablation study of key Components on the GD@Top5 and HD@Top5, which indicates the percentages of GD and HD in the top five documents. $-I(d)$ ignores $p(d_{gd} | q, d)$ in U' , and $-R(d)$ ignores $p'(d_{hd} | q, d)$. and the baseline represents the best result among our compared baseline approaches.

Components	MuSiQue		NQ		MultiHopQA		PQA-L	
	GD	HD	GD	HD	GD	HD	GD	HD
$-I(d)$	4.26	0.57	2.47	2.20	6.39	2.93	4.99	1.84
$-R(d)$	86.7	4.47	68.4	17.5	62.1	16.7	65.8	19.4
Ours	82.5	2.56	68.1	10.3	61.7	4.98	65.2	8.19
baseline	80.1	10.9	70.2	19.6	62.8	15.2	66.9	19.3

To further investigate the impact of $I(d)$ and $R(d)$ on the utility function, we further evaluate changes in the proportion of HD and GD for the reranked documents after ablation (see Tb. 4). $-I(d)$ fails to capture GD. This suggests that $I(d)$ functionally aligns with traditional reranker (Pearson correlation coefficient $\rho > 0.95$ compared to baseline), essentially enhancing LLM answer accuracy by increasing the proportion of GD. Introducing $R(d)$ sacrifices a small reduction in GD percentage while achieving substantial HD reduction, indicating that $R(d)$ improves overall performance by reducing the risk of LLMs being misled by reranked documents. At the same time, we observe that when $-I(d)$ (which implies $U(d|q) = -\lambda R(x)$), the utility function can still identify a portion of the ground-truth documents. We speculate that although $R(x)$ is designed to analyze document risk, the ground-truth documents are also included as risk-free samples in the training set of the FFN_{hd} , and it enables $R(d)$ to retain a certain ability to capture the characteristics of ground-truth documents.

Table 5: Computational Cost (Avg) of each baseline.

Method	BGE	JINA	GTE	ListCR	Ours
Computational Cost (Avg)	1.02	1.17	1.20	2.08	1

B Computational Cost

We compare the computational time cost between our method and baseline methods. The model definition and inference follow implementations from the original papers. Some baseline methods contain potential optimizations, such as JINA and BGE. As shown in the Tb. 5 We use our method as the baseline and report relative computational costs for each method.

C Datasets

C.1 Details of Datasets

MuSiQue is a multihop reading-comprehension QA dataset comprising approximately 50k QA pairs, with the test set consisting of around 2.5k samples. The dataset covers six reasoning-graph shapes spanning 2- to 4-hop questions and includes diverse knowledge types such as locations, dates, and people. Gold answers are typically unique and consist of a single word or phrase. We conduct our empirical study on the training set and evaluate our method on the test set.

Natural Questions (NQ) is an open-book QA dataset comprising approximately 323k QA pairs, with the test set consisting of around 7.8k samples. The dataset covers a broad range of knowledge types, including factual information about people, places, events, and concepts. The gold answers are typically unique and consist of a single word, phrase, or a short span of text. We evaluate our method on the test set.

MultiHop-RAG is an open-book dataset comprising approximately 2,556 multi-hop question-answer pairs. The dataset consists of four subsets covering query types such as inference, comparison, temporal, and null queries. The gold answers are typically unique and consist of a single word, entity name, or phrase. We evaluate our method on the entire dataset.

PubMedQA (PQA) is a biomedical research question answering dataset comprising approximately 213.3k QA pairs, with the test set consisting of around 500 samples. The dataset is split into three subsets: PQA-A (artificially generated), PQA-U (unlabeled), and PQA-L (labeled). It covers a wide range of biomedical topics, including clinical studies, treatment outcomes, and risk factors. The questions are derived from PubMed article titles and require reasoning over the corresponding abstracts to answer with yes/no/maybe. The gold answers are typically unique and consist of a single word (yes, no, or maybe). We evaluate our method on the entire PQA-L dataset.

C.2 Dataset Construction for Document Role Discriminator Training

First, we extract all sentences labeled as ground truth from the dataset. Subsequently, we vectorize all reference documents to construct a reference database. Using the MIPS (Maximum Inner Product Search) method, we retrieve the top-10 recalls. These results are then compared against the ground truth documents. We instruct GPT-4o to annotate all non-ground-truth documents (prompt can be

Table 6: p -values under the McNemar test between our method and the top-performing baseline in the main experiment (see Sec. Overall Performance).

Model	MuSiQue	NQ	MultiHopQA	PQA-L
GPT4o-mini	1.69e-16	2.45e-2	4.19e-3	-
gemma (4B)	4.13e-5	2.48e-2	1.70e-2	-
Qwen3 (8B)	1.72e-18	3.69e-4	7.66e-3	-

Table 7: p -values under the McNemar test between MIPS and each reranker in the empirical study.

Reranker	BGE	JINA	GTE	ListCR
p	2.17e-5	3.10e-12	6.74e-5	4.27e-19

found in our github repository), thereby constructing corresponding QA pairs. To validate its reliability, we split the dataset into training, validation, and test sets in an 8:1:1 ratio. On the test set, the FFN_{gd} achieves 94.2% accuracy, and the FFN_{hd} achieves 86.1%.

C.2.1 The definition of HD. We define four annotation label categories based on our corpus characteristics and Wu et al. [54]:

- **Semantic Noise (SeN):** Off-topic/irrelevant content
- **Illegal Sentence Noise (ISN):** Grammatically incoherent fragments
- **Counterfactual Noise (CN):** Factually incorrect claims
- **Supportive Noise (SuN):** Exhibit high semantic relevance to the query topic but lack specific answer details (including similar-yet-distinct subjects)

We classify CN and SuN as HD.

D Significance Test Results

We apply the McNemar test to evaluate whether our proposed method shows statistically significant improvement over baseline approaches. We also apply it to prove that the proportion of HD increases significantly after reranking. The McNemar test is a non-parametric statistical procedure used to analyze matched categorical data, especially useful for comparing two classification models' performance on identical data. It assesses whether differences in prediction disagreements between models are statistically significant. Tb. 6 shows that our improvement is significant ($p < 0.05$). Also, Tb. 7 suggests that the reranker significantly increases the proportion of HD.

E Additional Experiments on Alternative Retrievers

To verify that our conclusions on Sec. 3 are not model-specific, we extend our analysis to two other advanced retrievers: Qwen3-Embedding-0.6B [67] and jina-embeddings-v3 [46]. We follow the exact same experimental pipeline as detailed in Sec. 3 for a fair comparison. And the Fig. 5 provides strong evidence for the generalizability of our findings. We both observe that an increased proportion of GD alongside a significant rise in HD. This replication reinforces our central claim that current reranking models,

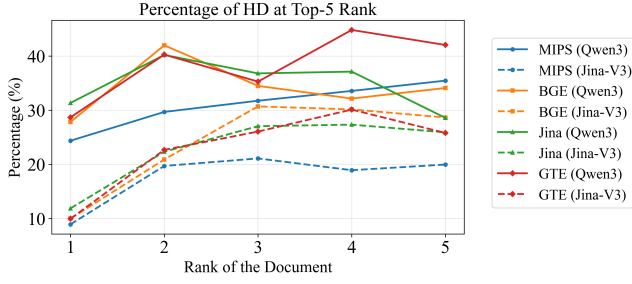


Figure 5: A comparison of reranking performance across different retrievers. We observe a consistent trend across Qwen3-Embedding-0.6B and jina-embeddings-v3, similar to our main results.

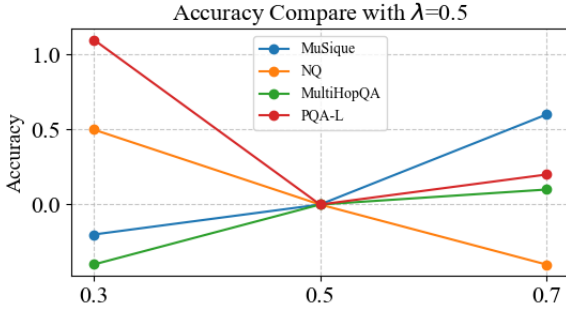


Figure 6: Accuracy variation across λ with $\lambda = 0.5$ as baseline.

while improving relevance, also inadvertently increase the risk of introducing high-ranked misinformation. This phenomenon persists across different foundational retrieval systems.

F Effect of λ

The parameter λ denotes the weighting coefficient for $R(d)$ in the utility function, representing risk sensitivity. We set $\lambda = 0.5$ and report its overall performance in our main experiment. In this section, we also report λ 's impact on accuracy in Fig. 6. First, λ does not substantially alter system effectiveness. Second, $\lambda = 0.5$ serves as a balanced choice. Increasing λ slightly improves performance on complex multi-hop problems but slightly reduces it on simple single-hop problems.

G Discussion of Clip

We select 0.5 as the threshold for Clip primarily due to the balanced distribution of our training data, where this value naturally serves as the optimal decision boundary between risk and safety categories. We find that threshold selection between 0.3 and 0.7 negligibly affects accuracy. However, thresholds exceeding 0.7 cause the component to progressively exhibit behavior similar to $-R(d)$, leading to performance degradation. In contrast, thresholds below 0.3 undermine the fundamental design objective of maintaining numerical stability in logarithmic computations, which similarly degrades performance.

H Different Performance Between Reasoning Models and Non-reasoning Models.

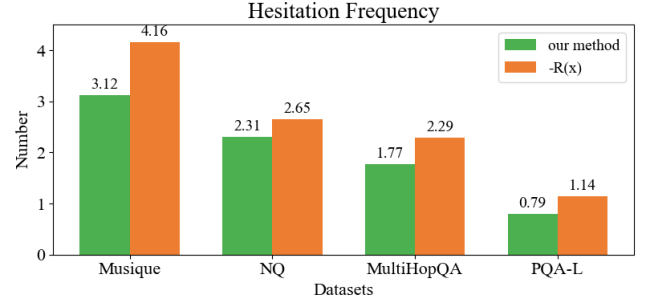


Figure 7: Hesitation frequency variation in reasoning models under the omission of $R(d)$ over three datasets. The y-axis shows hesitation word frequency.

To investigate why reasoning models exhibit greater performance sensitivity when we omit $R(d)$ (Sec. 5.3), we conduct an in-depth analysis of their output behavior. Specifically, we construct a set of keywords that indicate hesitation-expressing and compute their average frequency in the model outputs (see details in App. I). Our analysis reveals two key findings (Fig. 7). First, after omitting $R(d)$, reasoning models show a marked increase in hesitation. Second, this effect is significantly more pronounced for multi-hop questions than for single-hop ones. The findings suggest that HDs, which $R(d)$ is designed to mitigate, introduce greater confusion in reasoning models. This confusion likely diverts their reasoning paths toward unproductive paths, ultimately leading to the observed performance degradation (see the report of the proportion of GD and HD in App. A).

I Hesitation Word

We collect hesitation-indicating vocabulary from LLM reasoning traces to quantify hesitation frequency, with the lexicon comprising: *Alternatively, Wait, Rethink, I think I made a mistake, Reconsider, Hmm, Hold on*, etc.

J Limitations

Given the scarcity of open-source datasets, we have no choice but to employ GPT-4o for automatic annotation, a methodology widely adopted in both industry and academia. To ensure the fidelity of our data, we conduct a validation study with human evaluators and another LLM, achieving an “Almost Perfect” Kappa Coefficient. This high agreement confirms that any potential deviations from manual annotation are slight and well within acceptable bounds for research.

K Future Work

We will strengthen the theoretical and empirical analysis of why rerankers fail—particularly the “shortcut” hypothesis—through more rigorous probing studies and attention diagnostics. We also plan to explore unsupervised annotation, adaptive risk-aversion mechanisms, and robustness against adversarial attacks.